NXB Ledger CTF

Challenge descriptions & theme CSS — organizer reference

*This document contains the six player-facing challenge descriptions (ready to paste into CTFd) and the custom challenge-board CSS. Descriptions were reconstructed from the challenge specifications — please verify the wording against your live CTFd before publishing. The CSS below is the custom additions block; the complete **theme/ctfd-theme.css** file (including the base red/black theme) lives in the private GitHub repo and on the original CTF box — that file remains the source of truth.*

Challenge descriptions

Stage 1 — The Broker's Promise

OSINT · 100 pts

A loan application was leaked from the bank, and the trail starts with the broker who advertised it. Browse the broker classifieds site and find the advertisement matching the leaked loan: the ad promises **expedited processing** and charges a **documentation clearance fee of 2%**.

Identify the advertisement and submit its post ID, the broker's contact code, and the two distinguishing features to the evidence checker.

```
Broker site: http://127.0.0.1:8081
```

```
Evidence checker: http://127.0.0.1:8082/check/stage1
```

Stage 2 — The Hidden Conversation

Encoding / Scripting · 100 pts

Investigators intercepted a chat export related to the case. One of the messages isn't what it seems — its body is **encoded**. The attachment **chat_fragment.json** holds the message; decode the body field to reveal the hidden content, then submit it to the evidence checker.

```
Evidence checker: http://127.0.0.1:8082/check/stage2
```

Stage 3 — Beyond the Branch

Web Security · 200 pts

Log in to the bank's loan officer portal and investigate loan application **APP-0842**. The portal has an access control flaw: application records are addressed by predictable IDs, and nothing verifies that you are allowed to view another branch's files.

Find a way to open the target application and recover the flag hidden inside its file.

```
Portal: http://127.0.0.1:8080
```

```
Login: br03_officer / Training#2026
```

Stage 4 — The Workstation Trace

Digital Forensics · 200 pts

A screenshot of the leaked loan application was captured from a bank workstation. Correlate the forensic evidence to determine exactly **which workstation** the screenshot was taken from, and **which capture event** recorded it.

Evidence: asset_register.csv, capture_events.csv, leaked_screenshot_master.png, manifest_sha256.txt

Submit **workstation_id** and **capture_event** to the evidence checker.

```
Evidence checker: http://127.0.0.1:8082/check/stage4
```

Stage 5 — The False Trail

Log Analysis · 300 pts

Someone accessed the leaked application from the compromised workstation — but the logs are noisy, and one trail in the evidence is **deliberately false**. Reconstruct the true timeline: find the real session, identify the employee behind it, and name the unrelated failed event that should be discarded.

Evidence: auth.csv, app_access.csv, workstation_sessions.csv, leave.csv, alert_note.txt

Submit **session_id**, **employee_id** and **unrelated_failed_event** to the evidence checker.

```
Evidence checker: http://127.0.0.1:8082/check/stage5
```

Stage 6 — Reconstruct the Ledger

Programming / Scripting · 400 pts

Pull every thread together. A broker chat message references a payment tied to the leaked application. Write a script (**correlate.py**) that joins the chat messages, the staff directory, the broker ledger, the application access logs and the workstation sessions to reconstruct the complete evidence chain.

Evidence: messages.jsonl, directory.csv, broker_ledger.csv — plus Stage 5's app_access.csv and workstation_sessions.csv

Your script must output all six fields: **employee_id**, **application_id**, **workstation_id**, **session_id**, **broker_transaction_ref**, **message_id**. Submit them to the evidence checker.

```
Evidence checker: http://127.0.0.1:8082/check/stage6
```

Theme CSS — challenge-board custom block

*Append this block to **theme/ctfd-theme.css** (it is already appended in the repo's version of the file), then apply with:*

```
python3 ctfd-setup/apply_theme.py --url http://127.0.0.1:8000 --token YOUR_TOKEN_HERE
```

```
/* ============ NXB challenge board v6 - clean hacker look ============ */
@keyframes nxb-flicker {
  0%, 100% { opacity: 1; }
  8% { opacity: .35; } 9% { opacity: 1; }
  55% { opacity: .45; } 56% { opacity: 1; }
}
[x-data=&quot;ChallengeBoard&quot;]::before {
  content: &quot;NXB LEDGER&quot;;
  display: block;
  text-align: center;
  font-family: &quot;Courier New&quot;, Courier, monospace;
  font-size: clamp(2rem, 6vw, 3rem);
  font-weight: bold;
  letter-spacing: .3em;
  text-indent: .3em;
  color: #ff2a2a;
  padding: 2rem 0 .4rem;
  text-shadow: 0 0 18px rgba(255,42,42,.8), 0 0 70px rgba(255,42,42,.45);
  animation: nxb-flicker 6s infinite;
}
.jumbotron h1 {
  font-family: &quot;Courier New&quot;, Courier, monospace;
  letter-spacing: .25em;
  text-indent: .25em;
  text-shadow: 0 0 16px rgba(255,42,42,.55);
}
[x-data=&quot;ChallengeBoard&quot;] h3 {
  color: #ff3b3b !important;
  font-family: &quot;Courier New&quot;, Courier, monospace !important;
  letter-spacing: .22em;
  text-transform: uppercase;
  font-size: 1.05rem;
  text-shadow: 0 0 12px rgba(255,42,42,.6);
}
[x-data=&quot;ChallengeBoard&quot;] h3::before {
  content: &quot;&gt;_ &quot;;
  color: #ff8080;
}
[x-data=&quot;ChallengeBoard&quot;] .challenges-row &gt; div {
  flex: 0 0 auto;
  width: min(400px, 100%);
}
.challenge-button {
  position: relative;
  transition: transform .18s ease, box-shadow .18s ease, border-color .18s ease;
}
.challenge-button:hover {
  transform: translateY(-4px);
  border-color: #ff6b6b !important;
  box-shadow: 0 0 26px rgba(255,42,42,.65), inset 0 0 26px rgba(255,42,42,.12) !important;
}
.challenge-button span {
  color: #ff6b6b !important;
  font-family: &quot;Courier New&quot;, Courier, monospace;
  letter-spacing: .2em;
}
.challenge-button.challenge-solved::after {
```

```
  content: &quot;[\2713 SOLVED]&quot;;
  position: absolute;
  top: 7px; right: 9px;
  color: #22ff88;
  font-size: 10px;
  font-family: &quot;Courier New&quot;, Courier, monospace;
  letter-spacing: 1px;
}
body:has([x-data=&quot;ChallengeBoard&quot;])::after {
  content: &quot;&quot;;
  position: fixed;
  inset: 0;
  z-index: 9999;
  pointer-events: none;
  background: repeating-linear-gradient(to bottom, transparent 0px, transparent 3px, rgba(0,0,0,.12) 4px);
}
/* single hero: hide default jumbotron on challenges page, tagline under banner */
body:has([x-data=&quot;ChallengeBoard&quot;]) .jumbotron {
  display: none;
}
[x-data=&quot;ChallengeBoard&quot;] &gt; [x-show=&quot;loaded&quot;]::before {
  content: &quot;&gt; trace the money // 6 stages // 1300 pts&quot;;
  display: block;
  text-align: center;
  font-family: &quot;Courier New&quot;, Courier, monospace;
  font-size: .8rem;
  letter-spacing: .35em;
  text-indent: .35em;
  color: #ff8080;
  padding-bottom: 1.4rem;
  margin-bottom: 1rem;
  border-bottom: 1px solid rgba(255,42,42,.35);
}
/* full-width stage cards (replaces the 400px card width) */
[x-data=&quot;ChallengeBoard&quot;] .challenges-row{margin:0}
[x-data=&quot;ChallengeBoard&quot;] .challenges-row &gt; div{width:100%;max-width:none;flex:0 0 100%;padding
```